Sequences & Series

Arithmetic sequences: finding terms

1 Find the 12th term of the arithmetic sequence with $a = 5$, $d = 3$.

2 Find the 20th term of the arithmetic sequence with $a = -4$, $d = 2$.

3 Find the 15th term of the arithmetic sequence with $a = 100$, $d = -6$.

4 Write the first four terms of the arithmetic sequence with $a = 7$, $d = -2$.

Arithmetic sequences: finding a and d

5 The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find the first term a and common difference d.

6 An arithmetic sequence has $a_1 = 10$ and $a_6 = 30$. Find the common difference d.

7 An arithmetic sequence has $a_3 = 11$ and $a_7 = 27$. Find the common difference d.

Arithmetic series: the sum formula

8 Find the sum of the first 20 terms of the arithmetic sequence with $a = 3$, $d = 4$, using $S_n = \frac{n}{2}(2a + (n-1)d)$.

9 Find the sum of the first 15 terms of the arithmetic sequence with $a = 50$, $d = -3$.

10 Find the sum of the arithmetic series $2 + 5 + 8 + \cdots + 50$.

11 Find the sum of the first 10 positive multiples of 6.

Geometric sequences: finding terms

12 Find the 6th term of the geometric sequence with $a = 3$, $r = 2$.

13 Find the 5th term of the geometric sequence with $a = 100$, $r = 0.5$.

14 Find the 4th term of the geometric sequence with $a = 2$, $r = -3$.

15 Write the first four terms of the geometric sequence with $a = 5$, $r = 3$.

Geometric sequences: finding a and r

16 The 2nd term of a geometric sequence is 6 and the 5th term is 162. Find r and a.

17 A geometric sequence has $a_1 = 4$ and $a_4 = 108$. Find r.

18 A geometric sequence has 3rd term 12 and 6th term 96. Find a and r.

Geometric series: the finite sum

19 Find the sum of the first 6 terms of the geometric series with $a = 3$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

20 Find the sum of the first 5 terms of the geometric series with $a = 100$, $r = 0.5$.

21 Find the sum of the first 4 terms of the geometric series with $a = 2$, $r = -3$.

Infinite geometric series

22 Find the sum of the infinite geometric series with $a = 4$, $r = \frac{1}{2}$.

23 Find the sum of the infinite geometric series with $a = 9$, $r = \frac{1}{3}$.

24 Find the sum of the infinite geometric series $8 - 4 + 2 - 1 + \cdots$.

25 Express the repeating decimal $0.333\ldots$ as a fraction using an infinite geometric series with $a = \frac{3}{10}$, $r = \frac{1}{10}$.

Sigma notation

26 Evaluate $\sum_{k=1}^{5} (2k + 1)$.

27 Evaluate $\sum_{k=1}^{4} 3^k$.

28 Use $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{10} k$.

Recursive sequences

29 A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n + 5$. Find a_5.

30 A sequence satisfies $a_1 = 3$ and $a_{n+1} = 2a_n - 1$. Find a_4.

31 A sequence satisfies $a_1 = 1$, $a_2 = 1$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. Find a_6.

Identifying arithmetic and geometric sequences

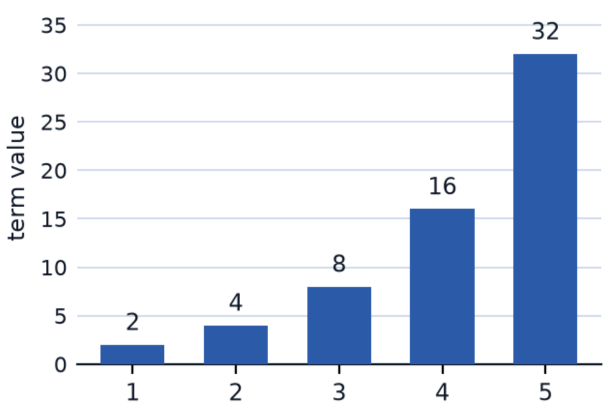
32 State whether the sequence $4, 9, 14, 19, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

33 State whether the sequence $2, 6, 18, 54, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

34 State whether the sequence $100, 90, 80, 70, \ldots$ is arithmetic or geometric, and find its 10th term.

Reading a sequence from its terms

35 The bar chart shows the first five terms of a sequence.



a Determine whether the sequence is arithmetic or geometric, and find the common ratio or difference.

b Find the 8th term.

Word problems: sequences in context

36 This question has two parts. An employee's starting monthly salary is 4000 dollars, and it increases by a fixed amount of 150 dollars each month.

a Write a formula for the salary a_n in month n (with $a_1 = 4000$).

b Find the salary in month 12.

37 This question has two parts. A bacteria culture starts with 500 cells and triples every hour. Let a_n be the number of cells after n hours, with $a_0 = 500$.

a Write a formula for a_n.

b Find the number of cells after 4 hours.